

ROVANIEMI APUKKA, Finland

WMO: 028130

Lat: 66.579N

Lon: 26.011E

Elev: 352

StdP: 14.51

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.6	-20.6	-32.6	0.9	-26.2	-26.1	1.3	-20.3	16.1	27.4	14.2	24.4	1.1	290	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	77.9	62.5	74.0	60.9	70.5	59.3	65.6	73.7	63.3	71.1	61.0	67.5	5.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.6	85.9	69.6	60.1	78.7	66.0	58.2	73.3	63.9	30.6	74.0	28.9	71.5	27.3	67.7	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.5	11.2	-31.9	80.5	6.8	4.3	-36.8	83.6	-40.7	86.1	-44.5	88.5	-49.4	91.6		
(5)			-31.0	68.2	5.6	2.4	-35.1	70.0	-38.4	71.4	-41.5	72.8	-45.6	74.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.7	8.5	11.6	19.4	31.4	44.3	52.9	59.2	54.6	46.5	33.6	24.4	16.5		
	DBStd	19.49	13.17	15.41	10.83	7.67	7.91	6.96	5.87	6.01	5.91	7.38	10.29	12.89		
	HDD50	6608	1285	1074	948	559	215	47	3	25	135	510	768	1039		
	HDD65	11450	1750	1494	1413	1009	642	369	195	327	554	975	1218	1504		
	CDD50	655	0	0	0	0	38	133	288	166	30	0	0	0		
	CDD65	25	0	0	0	0	0	6	15	4	0	0	0	0		
	CDH74	336	0	0	0	0	19	63	211	43	0	0	0	0		
	CDH80	52	0	0	0	0	1	11	35	5	0	0	0	0		
	WSAvg	5.0	3.8	4.4	5.4	5.5	6.2	6.1	5.2	5.0	4.9	4.9	4.4	4.4		
	PrecAvg	21.1	1.5	1.2	1.2	1.1	1.3	2.0	2.4	2.8	2.4	2.1	1.8	1.4		
(15) Precipitation	PrecMax	28.5	3.7	2.9	2.9	3.3	3.5	4.5	5.2	5.5	4.8	4.3	3.4	2.3		
	PrecMin	15.5	0.4	0.4	0.1	0.2	0.2	0.3	0.5	0.7	0.3	1.0	0.5	0.3		
	PrecStd	3.3	0.7	0.7	0.7	0.7	0.7	1.1	1.3	1.2	1.1	0.9	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.8	39.7	43.2	59.6	76.5	80.8	84.2	80.2	65.2	50.8	42.6	37.1		
		MCWB	33.3	36.5	36.6	46.4	56.1	63.5	67.8	66.9	57.1	48.0	42.0	36.2		
	2%	DB	31.9	35.3	39.3	50.3	71.3	74.4	79.7	73.3	62.0	48.3	40.3	34.8		
		MCWB	31.4	33.6	34.0	40.8	56.4	60.4	63.6	61.1	54.8	45.5	39.3	33.9		
	5%	DB	29.9	33.1	36.5	46.3	64.9	70.1	75.5	69.3	59.4	46.2	37.7	33.3		
		MCWB	29.4	32.2	32.7	38.8	52.0	57.6	62.3	59.5	53.7	44.3	37.1	32.8		
	10%	DB	27.1	31.1	34.1	42.8	59.5	66.1	71.9	65.8	56.6	44.0	35.5	32.3		
		MCWB	26.4	30.4	31.9	36.6	49.4	55.2	61.0	58.2	52.7	42.5	35.0	31.8		
	0.4%	WB	33.6	36.8	37.6	46.7	58.9	65.4	69.0	67.8	58.7	49.4	42.1	36.5		
		MCDB	34.3	39.0	42.2	57.9	71.4	75.2	77.4	77.9	62.8	50.1	42.2	37.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.5	34.0	35.1	42.2	56.2	62.7	66.5	63.9	56.4	46.1	39.3	34.1		
		MCDB	31.9	34.7	38.1	48.9	69.4	73.3	75.0	70.0	59.9	47.3	39.9	34.4		
	5%	WB	29.3	32.4	33.5	39.4	53.6	60.0	64.6	61.2	54.8	44.3	37.1	32.9		
		MCDB	29.7	33.2	35.4	45.1	62.9	66.7	72.0	66.1	58.1	45.9	37.6	33.3		
	10%	WB	26.4	30.3	32.2	37.4	50.3	57.1	62.4	59.3	53.1	42.5	35.1	31.8		
		MCDB	27.1	31.0	33.9	41.6	57.7	62.9	70.2	63.8	55.8	44.0	35.6	32.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	16.4	20.9	17.7	17.9	16.4	17.6	16.7	13.9	9.9	9.9	13.4		
		MCDBR	17.2	10.7	17.7	22.6	29.0	24.3	25.6	23.4	20.0	12.5	8.1	10.7		
		MCWBR	17.5	9.8	13.7	14.2	14.7	11.4	12.5	13.1	13.3	10.4	8.0	10.6		
		MCDBR	17.3	10.2	12.1	20.1	26.0	19.9	21.7	18.5	15.2	11.2	8.1	10.2		
	5% WB	MCWBR	17.6	9.4	10.0	13.1	14.0	10.6	11.7	11.4	10.9	9.8	8.0	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.197	0.265	0.298	0.332	0.344	0.345	0.359	0.342	0.317	0.289	0.347	N/A		
	taud		1.933	2.103	2.164	2.238	2.445	2.471	2.450	2.481	2.533	2.345	2.655	N/A		
	Ebn at Noon		63	180	238	261	272	274	266	259	239	177	68	0		
	Edh at Noon		5	17	27	33	30	30	30	26	19	14	5	0		
(44) All-Sky Solar Radiation	RadAvg		22	177	634	1176	1502	1637	1531	1113	584	201	41	0		
	RadStd		2	32	65	104	134	122	148	102	61	39	5	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	+1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page